

Campus Energy Audit Report 2025

EXECUTIVE SUMMARY

This report documents the annual energy consumption and comprehensive utility audit for the Academic and Residential blocks of the main university campus for the fiscal year 2025. The campus serves approximately 2,000 students and 300 staff members. The audit was conducted to identify saving opportunities, assess renewable energy integration, and map operations to UN SDG 7 (Clean Energy) and SDG 13 (Climate Action).

1. KEY CONSUMPTION DATA

Total annual grid electricity consumption was evaluated at 520,000 kWh, with peak demand of 115 kW occurring during the summer months of April and May. The average monthly utility expenditure stands at INR 3,46,000, totaling INR 41,52,000 annually. Currently, the campus has 0 kW of installed on-site renewable energy, leaving it 100% dependent on the carbon-intensive local thermal power grid.

2. IDENTIFIED INEFFICIENCIES & ISSUES

1. HVAC System Setpoints: The air conditioning units in the main computer engineering labs and administrative server rooms are set at 18 degrees Celsius (64 degrees Fahrenheit) continuously. They run 24 hours a day, including weekends when student labs are closed.
2. Obsolete Lighting Infrastructure: Over 800 standard T8 fluorescent tubes (40W each) remain in service in the hostel corridors and library reading rooms. These units are active throughout the night.
3. Canteen Ventilation: Kitchen exhaust systems and industrial ventilation fans are left running at full power overnight when the canteens are completely closed.

3. RECOMMENDATIONS & SDG MAPPING

- Retrofit all 800 fluorescent tubes with 18W energy-efficient LED tubes. This is expected to save 38,500 kWh annually.
- Reset all classroom and lab thermostat setpoints to a standardized 24 degrees Celsius. Enforce automatic shutdown policies via smart plugs after 6 PM.
- Feasibility Study: Install a 100 kWp rooftop solar photovoltaic array on the Academic Block roof. This will generate approximately 146,000 kWh of clean power annually, reducing grid dependency by 28% and directly advancing SDG 7 target 7.2.
- Implementing these actions will offset approximately 44 metric tons of CO2 equivalent emissions annually, supporting SDG 13 (Climate Action).